

FINANCIAL PERFORMANCE ANALYSIS REPORT

MyUSA Bank | 2022 to 2023

Profitability, efficiency, returns and stock performance

Python	pandas	SQLite	SQL	Matplotlib
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Portfolio analysis based on a simulated financial dataset.

Project Objectives

Purpose of the Project

The Financial Performance Analyser turns raw simulated banking data into an evidence based assessment of how the bank performed across profitability, interest generation, returns, operating efficiency and market performance between 2022 and 2023. It also demonstrates the complete analytical process from data cleaning and validation through SQL and pandas analysis, visualisation and communication of findings.

Objectives

- Clean and validate the raw dataset before analysis.
- Create and interpret financial measures for profitability, interest performance, returns and efficiency.
- Use SQL and pandas to identify trends, exceptional periods, yearly changes and relationships between variables.
- Compare 2022 with 2023 to identify where performance improved, weakened or remained stable.
- Communicate the evidence through clear visualisations, conclusions and stated limitations.

What Each Core Metric Tells Us

Metric	What it measures	What it tells us about the bank
Net income	Overall profit after income and costs are combined.	Whether overall profitability is strengthening or weakening.
Net interest income	Interest income less interest expense.	The absolute earnings generated from the core interest spread.
Net interest margin	Net interest income relative to average earning assets.	How effectively earning assets generate net interest income.
Return on assets	Net income relative to total assets.	How effectively the asset base generates profit.
Return on equity	Net income relative to shareholder equity.	The return generated from shareholder capital.
Cost to income ratio	Operating expenses relative to operating income.	Operating efficiency. Lower values indicate stronger efficiency.
Stock price	The simulated market price of the bank shares.	Whether market valuation moves alongside financial performance.

Together, the metrics provide a broader view of performance than any single measure can provide on its own.

Executive Summary

The analysis shows a mixed 2023 performance picture. Net interest margin improved, but broader profitability and returns weakened. Operating efficiency also deteriorated slightly, while average stock price remained almost unchanged.

Annual Performance Comparison

Metric	2022	2023	Direction
Net income	\$1.616m	\$1.562m	Down 3.29%
Net interest margin	3.05%	3.14%	Up 2.98%
Return on assets	1.80%	1.74%	Down 3.25%
Return on equity	6.61%	6.38%	Down 3.48%
Cost to income	27.91%	28.28%	Up 1.29%
Stock price	\$126.48	\$126.35	Down 0.10%

Key Findings

- Average net income decreased by 3.29% in 2023, while net interest margin increased by 2.98%.
- The stronger margin was supported by a 1.41% rise in average interest income and a 1.75% fall in average interest expense, which lifted average net interest income by 3.55%.
- Average operating expenses increased by 0.78% while operating income edged down by 0.08%, consistent with the cost to income ratio increasing from 27.91% to 28.28%.
- ROA and ROE both declined as net income fell while total assets were almost unchanged and shareholder equity increased slightly.
- Average stock price was effectively flat. Contemporaneous Pearson correlations with the tested financial metrics were negligible.
- No negative net income observations were present in the cleaned dataset.

Overall assessment: stronger interest spread performance did not translate into stronger overall profitability in 2023.

1. Data and Methodology

The project uses a simulated time series dataset for a fictional bank. The cleaned dataset contains 499 dated observations, with 250 observations in 2022 and 249 in 2023. The SQLite database contains the same 499 unique dates.

Data Quality

- Forty nine rows contain missing values in both interest income and net income.
- Net interest income and net interest margin are also missing in those rows because they depend on interest income.
- Return on assets and return on equity are missing in the same rows because they depend on net income.
- The missing values were not imputed. pandas and SQL therefore calculate averages using the available observations for each metric.
- Potential outliers were identified using the IQR method and retained rather than automatically removed.
- No duplicate dates are present in the final SQLite table.

Calculated Metrics

Metric	Calculation
Net interest income	Interest income less interest expense
Net interest margin	Net interest income / Average earning assets x 100
Return on assets	Net income / Total assets x 100
Return on equity	Net income / Shareholder equity x 100
Cost to income ratio	Operating expenses / Operating income x 100

Analytical Workflow

Python and pandas handle cleaning, validation and comparative analysis. SQLite stores the processed data. SQL identifies exceptional periods and performs aggregation. pandas calculates annual changes and Pearson correlations. Matplotlib is used for monthly trend visualisations.

Currency note: the source data does not explicitly state a currency. US dollar formatting is used only as a presentation convention for monetary values.

2. Profitability and Interest Margin

Average net income declined from \$1.616m in 2022 to \$1.562m in 2023. The lowest monthly average occurred in September 2023 at approximately \$1.406m, while the strongest monthly average occurred in July 2022 at approximately \$1.752m.

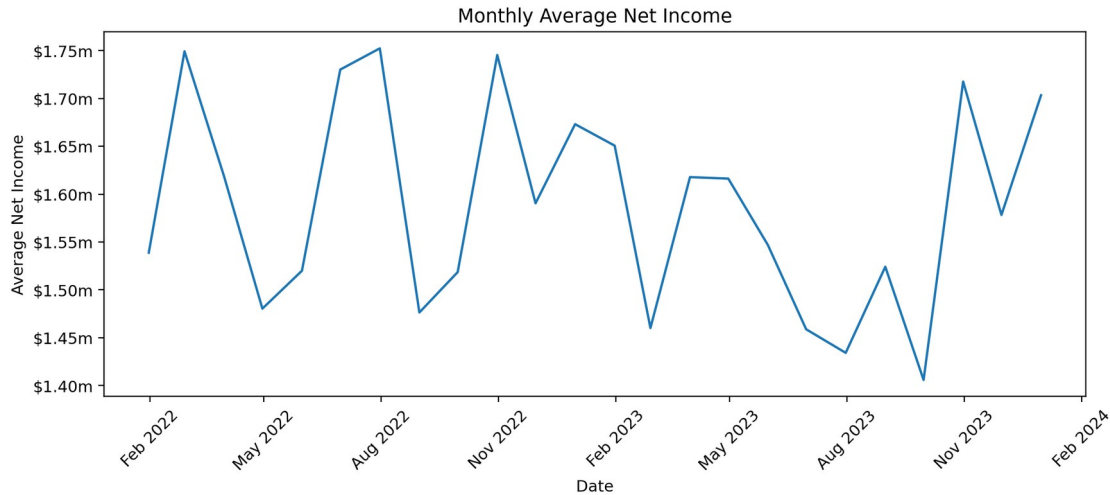


Figure 1. Monthly average net income.

Net interest margin moved in the opposite direction, increasing from 3.05% to 3.14%. The monthly series peaked at 3.42% in July 2023 and reached its lowest monthly average at 2.83% in June 2022.

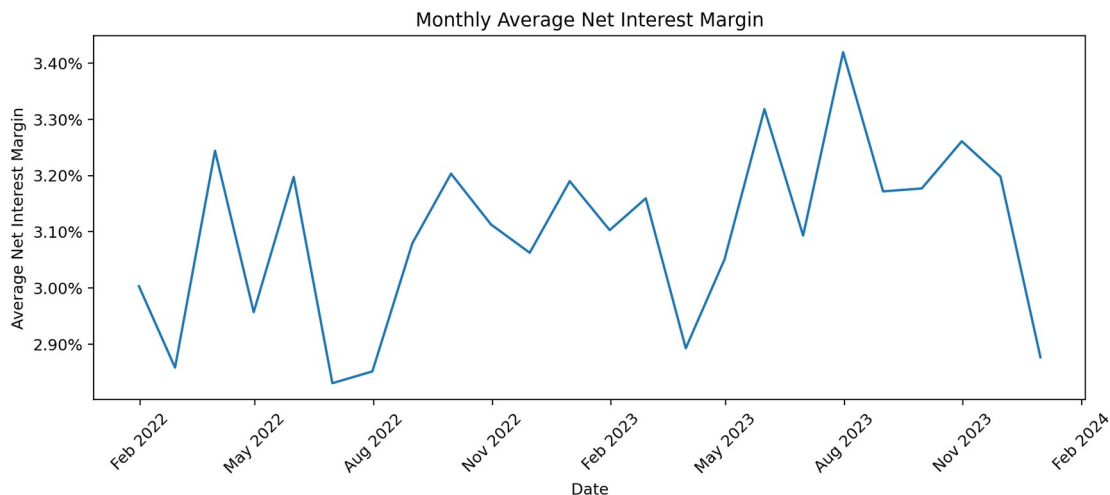


Figure 2. Monthly average net interest margin.

What Changed Underneath the Headline Metrics?

Average interest income increased by 1.41% while average interest expense decreased by 1.75%, lifting average net interest income by 3.55%. This is consistent with the improvement in net interest margin.

Average operating expenses increased by 0.78% while operating income edged down by 0.08%, which is consistent with the slight deterioration in cost efficiency. These movements do not fully explain the decline in net income because the dataset does not provide a complete income statement bridge, so no unsupported causal explanation is assigned.

3. Returns and Operating Efficiency

Average ROA declined from 1.80% to 1.74% and average ROE declined from 6.61% to 6.38%. Both monthly measures reached their lowest averages in September 2023, which aligns with the weakest monthly net income period.

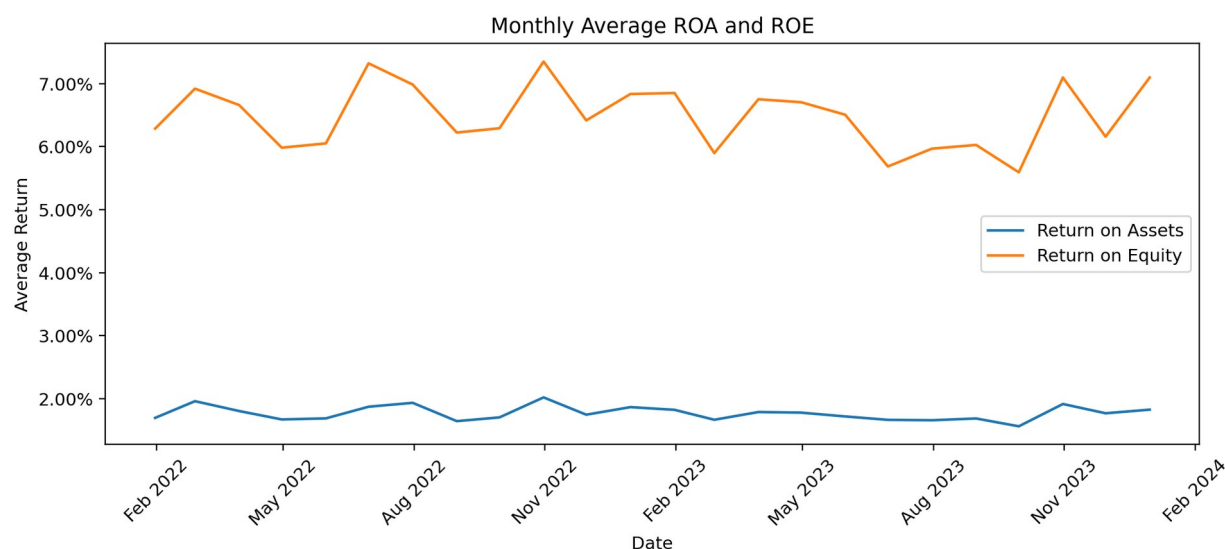


Figure 3. Monthly average return on assets and return on equity.

The decline in returns is consistent with the underlying balance sheet movements. Average total assets were essentially unchanged between the two years, while average shareholder equity increased by 0.40%. With average net income falling by 3.29%, both return measures were pressured downward.

Cost to Income Ratio

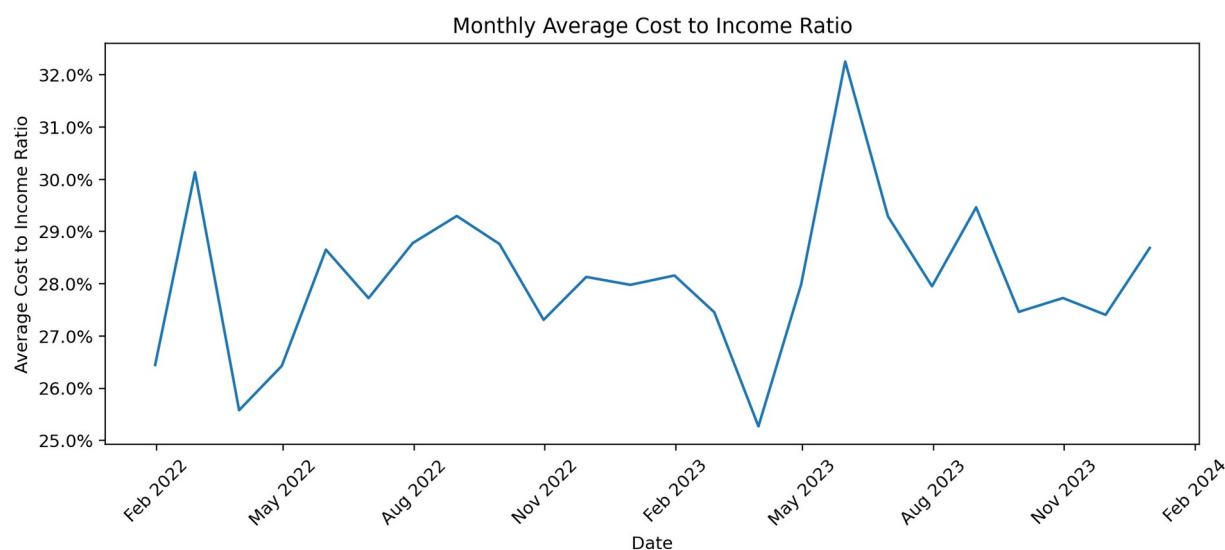


Figure 4. Monthly average cost to income ratio. Lower values indicate stronger efficiency.

The annual ratio increased from 27.91% to 28.28%. Monthly efficiency was volatile, with the best monthly average at 25.27% in March 2023 and the weakest at 32.25% in May 2023. The combination of slightly higher operating expenses and slightly lower operating income is consistent with this deterioration.

4. Stock Price and Financial Performance

Average stock price was almost unchanged, moving from \$126.48 in 2022 to \$126.35 in 2023. The monthly average peaked at approximately \$146.05 in December 2022 and reached its lowest point at approximately \$108.32 in February 2023.

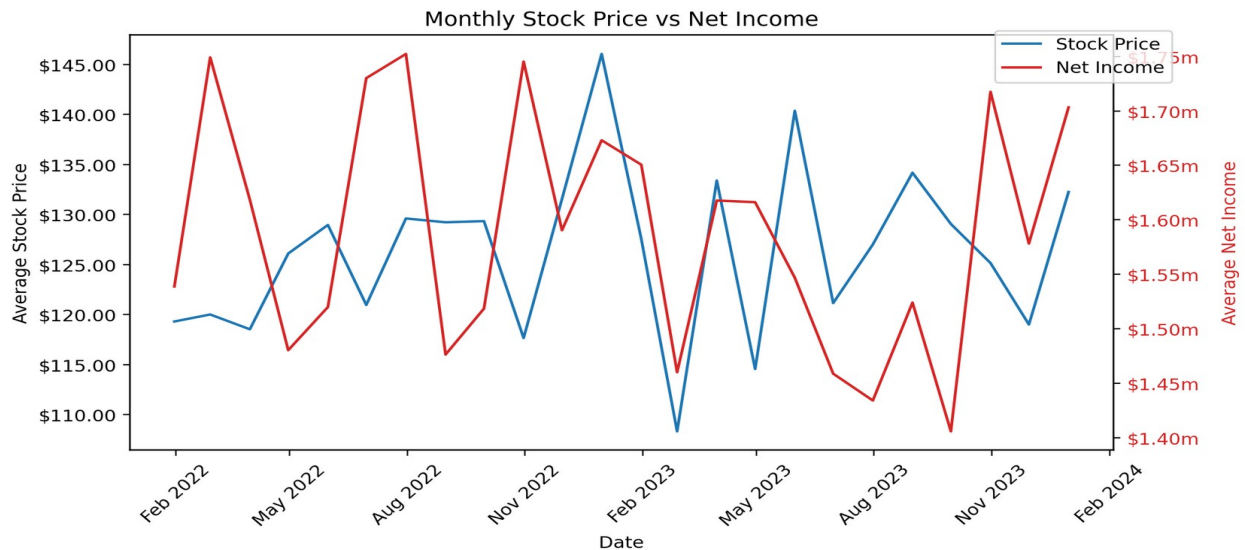


Figure 5. Monthly average stock price and monthly average net income shown on separate y axes. The chart is used for directional comparison only, while the Pearson analysis below is used to assess linear association.

Pearson Correlation Results

Metric vs Stock Price	Pearson r	Interpretation
Net income	+0.047	Negligible
Net interest margin	-0.011	Negligible
ROA	+0.040	Negligible
ROE	+0.077	Negligible
Cost to income	-0.029	Negligible

All coefficients are extremely close to zero. These are contemporaneous Pearson correlations across the available observations and should be treated as exploratory. The dataset covers only two years, and the analysis does not test lagged effects or nonlinear relationships. The results therefore support only the narrower conclusion that no meaningful contemporaneous linear association was detected.

5. Selected SQL Findings and Limitations

Selected SQL Extremes

The SQL stage identifies individual periods with exceptional values. These observations are useful for investigation but should not be treated as representative of the broader annual trend.

Query Result	Date	Value
Highest individual net income	11 Oct 2023	\$3.999m
Highest individual net interest margin	04 May 2023	4.75%
Lowest individual cost to income ratio	17 Oct 2023	14.22%

No negative net income observations were found in the cleaned database. The analysis therefore concerns changes in the level of profitability rather than a transition between profit and loss.

Limitations

- The dataset is simulated and does not represent the performance of a real financial institution.
- The source does not explicitly state a currency, so US dollar formatting is a presentation convention only.
- ROA uses total assets at each observation rather than average assets across a reporting period.
- ROE uses shareholder equity at each observation rather than average equity across a reporting period.
- Forty nine rows contain missing interest income and net income values. These values were not imputed.
- Monthly averaging improves readability but can conceal individual spikes and extreme observations.
- The dataset covers only two years, limiting longer term inference.
- Pearson correlation tests linear association only. Lagged and nonlinear relationships were not analysed.
- The dataset does not contain enough income statement detail to fully attribute the decline in net income to specific drivers.

6. Conclusion

The overall assessment is that 2023 was a mixed year for the simulated bank. The bank improved the economics of its core interest activities, but that improvement did not translate into stronger overall profit, returns or operating efficiency. The measures therefore need to be interpreted together rather than in isolation.

Metric by Metric Conclusion

Metric	What the result tells us about the bank
Net income	Average net income decreased by 3.29%, showing that overall profitability weakened in 2023.
Net interest income	Average net interest income increased by 3.55%, showing stronger absolute earnings from the interest spread.
Net interest margin	NIM increased from 3.05% to 3.14%, showing improved efficiency in generating net interest income from earning assets.
Return on assets	ROA decreased from 1.80% to 1.74%, showing that the bank generated slightly less profit from an almost unchanged asset base.
Return on equity	ROE decreased from 6.61% to 6.38%, showing a weaker return on shareholder capital as profit fell while equity increased slightly.
Cost to income ratio	The ratio increased from 27.91% to 28.28%, showing slightly weaker operating efficiency because more operating income was absorbed by operating costs.
Stock price	Average stock price was almost unchanged, showing that simulated market valuation remained broadly stable despite changes in the financial measures.

Supporting Financial Drivers

Average interest income increased by 1.41% and average interest expense decreased by 1.75%, supporting the improvement in net interest income and NIM. Average operating expenses increased by 0.78% while operating income decreased by 0.08%, supporting the weaker cost to income ratio. Total assets were almost unchanged and shareholder equity increased by 0.40%, so the declines in ROA and ROE are consistent primarily with the fall in net income rather than a major contraction in the balance sheet.

Taken together, the bank showed stronger interest spread performance but weaker overall profitability, returns and operating efficiency. The stable stock price and negligible contemporaneous correlations also show that the simulated market measure did not closely track the short term movements in the tested financial metrics. This mixed profile is the central conclusion of the analysis.

END OF REPORT